

Doorstep Car Wash Platform — Architecture

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Overview

Customers book a doorstep car wash online and pay the business directly; workers receive and complete the jobs; the admin controls cities, prices, offers and payouts from one panel.

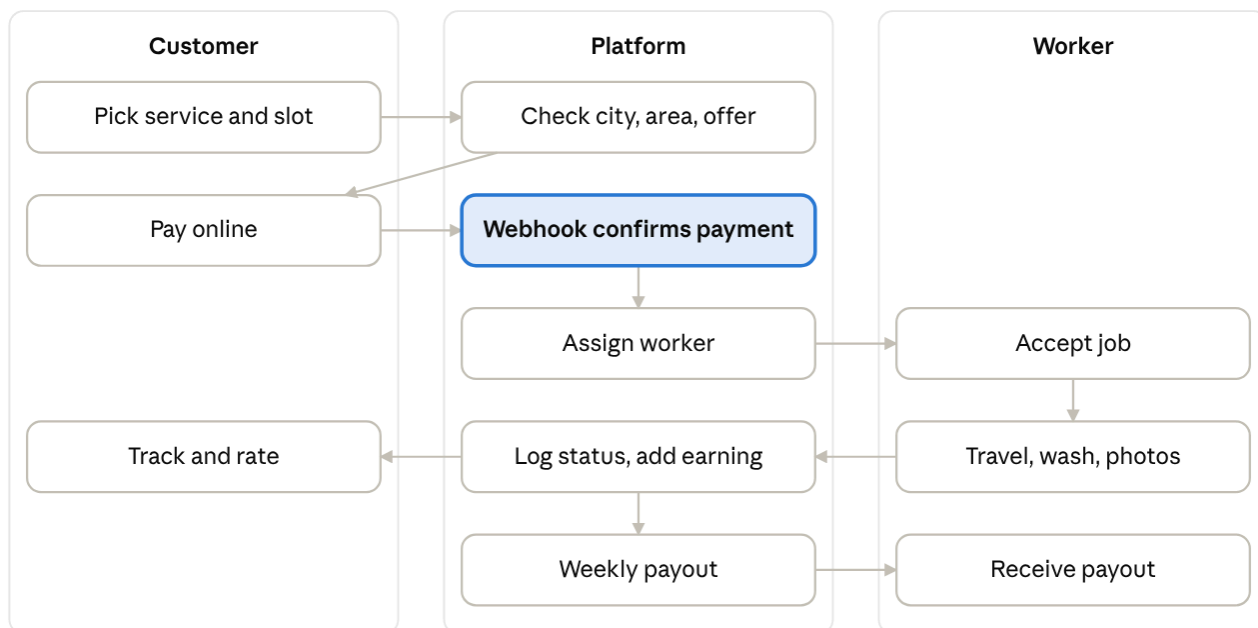
- **Three roles:** Customer (books and pays), Worker (does the wash), Admin (runs the business).
- **City-gated:** bookings work only in cities and areas the admin has switched on.
- **Money rule:** every payment goes to the business account, never to the worker. The system records each worker's earnings per job, and the admin pays workers on a schedule (e.g. weekly).
- **Admin-only control:** services, prices, slots, workers and all discounts are set from the admin panel, with no code changes.

Prices in this doc use ₹ and Razorpay; for a UK launch, swap to £ and Stripe.

How a booking flows

Customers pay the business; the platform assigns the job and pays workers

One booking from choice to payout, read top to bottom



booking journey · customer, platform, worker

The payment is confirmed only by the gateway's webhook, and the money settles in the business account; the worker's share is recorded per job and paid in the weekly payout.

Roles and features

All three roles use one web codebase: `/` for customers, `/worker` for workers (installable as a phone app), `/admin` for the business.

Role	App	Main features
Customer	Website (mobile app later)	OTP login, save cars and addresses, check city/area coverage, choose package, pick slot, apply offer, pay online, track worker, cancel or reschedule, booking history, invoices, ratings
Worker	Mobile web app (PWA)	Login, go online/offline, see assigned jobs, accept or reject, navigate to address, update status, upload before/after photos, record cash (if allowed), view earnings and payouts
Admin	Web dashboard	Cities and areas, services and prices, slots, workers, bookings and assignment, payments and refunds, worker payouts, offers, customers, content, reports, sub-admin users

City-based service control

Bookings are accepted only where the admin has switched a city on and the customer's pincode or GPS point falls inside an active service area.

Launching a city (admin): add city → draw service areas or add pincodes → set prices per service and vehicle type → set working hours and slots → assign workers → toggle city to Active.

What the customer sees:

Situation	Result
City active and address inside an active area	Services, prices and slots shown; booking allowed
City active but address outside all areas	"We don't serve your area yet" + Notify me
City inactive or not listed	"Coming soon to your city" + Notify me

Waitlist: "Notify me" stores the phone number, city and pincode. When the admin activates that city, the system sends every waitlisted person an SMS or WhatsApp message.

Server-side checks on every slot search and booking (the frontend alone is never trusted):

1. City is active.
2. Address is inside an active service area.
3. Service is active in this city; price is read from the city's price list.
4. A slot has spare capacity and a worker is available.

Switching a city off stops new bookings immediately; bookings already confirmed still go ahead.

Admin panel

Every business setting lives in the database and is edited from the admin panel, so launching a city, changing a price or running an offer needs no code change.

Module	What the admin controls
Cities	Add cities; activate or deactivate them
Service areas	Zones drawn on a map or listed by pincode; extra charge for far zones
Services and pricing	Packages (Basic, Premium, Interior, Bike); price per city and vehicle type; on/off
Slots	Working hours, slot length, capacity, holidays and blocked dates per city
Workers	Onboarding, KYC documents, city and area, commission, activate or suspend

Module	What the admin controls
Bookings	View and filter, assign or reassign, cancel, refund, notes
Payments	Transactions, refunds, cash collected, worker payouts
Offers	Create, edit, pause and track all discounts (see next section)
Customers	Profiles, booking history, block, wallet credit
Content	Homepage banners, FAQs, terms, refund policy, contact details
Reports	Revenue per city, bookings per day, worker performance, offer usage
Admin users	Sub-admins with limited access

Admin roles:

Role	Access
Super Admin (owner)	Everything: all cities, payments, offers, settings
City Manager	Own city's bookings, workers and slots only; cannot create offers or see total revenue
Support	View bookings, handle complaints, raise refund requests for Super Admin approval

Offers and discounts

Only the Super Admin creates or edits offers; customers can only apply offers that exist and pass every rule, and the discount is always calculated on the server.

Offer types:

Type	Example
Coupon code, flat	WASH50 : ₹50 off
Coupon code, percent	20% off, capped at ₹100
First booking	30% off the first wash, new users only

Type	Example
City launch	50% off in one city for the first 15 days
Service or vehicle specific	15% off Premium Wash, or SUV only
Auto-applied	Weekday discount applied without a code
Referral	₹100 wallet credit to both referrer and new user
Subscription plan	4 washes a month at a fixed price

Rules each offer can carry: discount type (flat or percent), value, maximum discount, minimum order value, cities, services, vehicle types, user type (all, new, specific), valid from/to, valid weekdays, total usage limit, per-user limit, auto-apply, active on/off.

How a code is applied:

1. Customer enters a code at checkout.
2. Server checks: offer active; today within dates and allowed weekdays; customer's city, service and vehicle type match; order at or above minimum; total and per-user limits not reached; new-user rule met.
3. Server calculates the discount and final amount, then creates the payment order for that amount.
4. On payment success, one usage row is recorded against the offer and customer.

The admin's offers screen lists each offer's code, discount, cities, expiry, usage against limit, and an on/off switch. Active offers can also show as homepage banners, filtered to the customer's city.

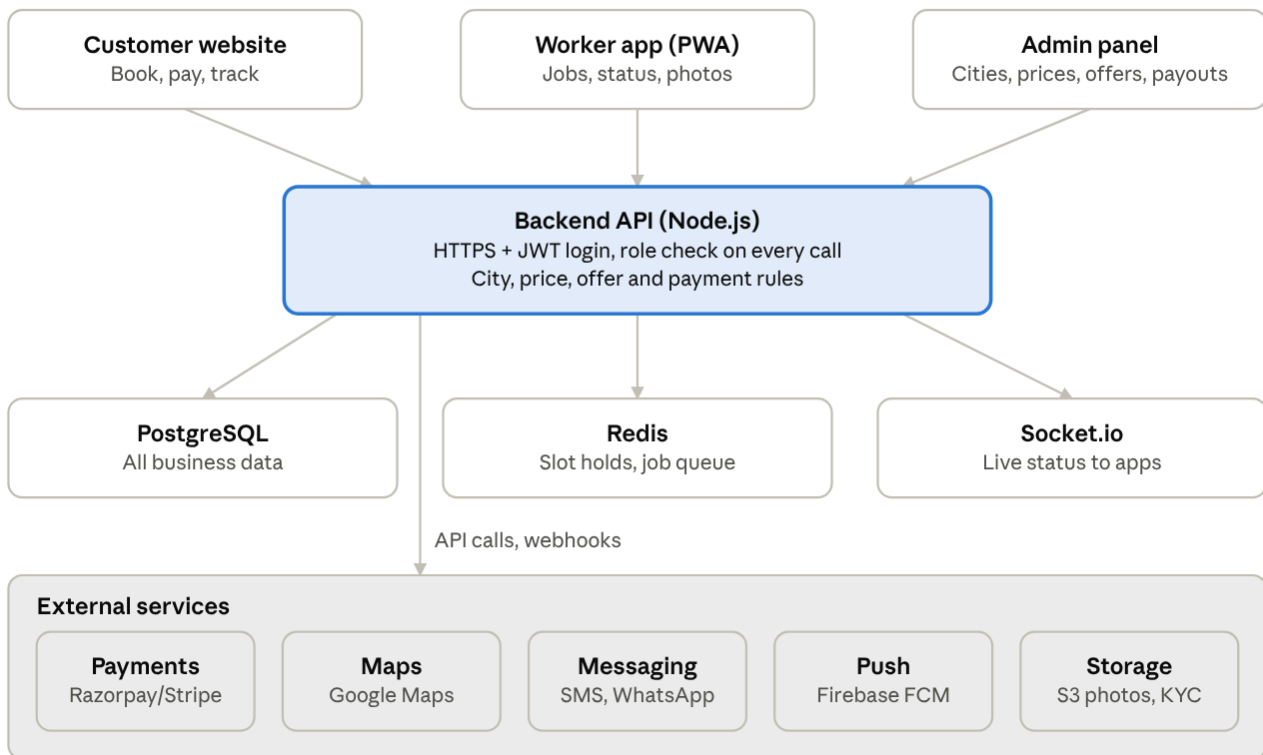
Tech stack and architecture

One Next.js frontend and one Node.js backend serve all three roles; the backend alone talks to the database and outside services.

Layer	Choice	Why
Frontend	Next.js (React)	One codebase for customer, worker and admin; worker part installable as PWA
Backend	Node.js (NestJS or Express), REST API	Business rules, auth, webhooks
Database	PostgreSQL	Relational data and transactions for bookings and money

Layer	Choice	Why
Cache and queue	Redis	Hold slots during payment; reminders and auto-assignment jobs
Payments	Razorpay (India) or Stripe (UK and others)	UPI/cards; worker payouts via Route or Connect
Maps	Google Maps API	Address picking, area check, distance, live tracking
Messaging	MSG91 or Twilio, WhatsApp Business API	OTP, confirmations, reminders
Push	Firebase Cloud Messaging	New-job alerts for workers
File storage	AWS S3 or Cloudinary	Before/after photos, KYC documents
Real-time	Socket.io	Live job status and tracking
Hosting	Vercel (frontend); AWS, DigitalOcean or Railway (backend, database)	Low cost to start, scales later

All three apps talk to one backend, which owns the data and every external service



system architecture · 3 apps, 1 backend

Customers, workers and admins never reach the database or payment gateway directly; every request passes the backend's login and role checks.

Database schema

PostgreSQL, grouped by area. Every booking status change is logged so disputes can be settled from history.

Area	Table	Key columns
Users	users	id, name, phone, email, role (customer, worker, admin), status
Users	addresses	id, user_id, label, address, lat, lng, pincode
Users	vehicles	id, user_id, type (hatchback, sedan, SUV, bike), brand, model, number
Cities	cities	id, name, state, is_active, launch_date, timezone, currency

Area	Table	Key columns
Cities	service_areas	id, city_id, name, pincodes or polygon, is_active, extra_charge
Cities	city_settings	city_id, open_time, close_time, slot_duration_min, min_notice_hrs, cancellation_window_hrs, max_bookings_per_slot
Cities	waitlist	id, phone, email, city_name, pincode, notified
Services	services	id, name, description, duration_min
Services	city_services	id, city_id, service_id, vehicle_type, price, is_active
Services	time_slots	id, area_id, date, start, end, capacity, booked_count
Workers	workers	user_id, city_id, area_id, is_online, rating, id_proof_url, bank_details, commission
Bookings	bookings	id, customer_id, vehicle_id, address_id, city_service_id, slot_id, worker_id, status, amount, discount, offer_id
Bookings	booking_status_log	booking_id, status, changed_by, timestamp
Bookings	booking_photos	booking_id, type (before, after), url
Bookings	reviews	booking_id, rating, comment
Money	payments	id, booking_id, gateway_order_id, gateway_payment_id, amount, method, status
Money	refunds	id, payment_id, amount, reason, status
Money	worker_earnings	id, worker_id, booking_id, amount, status (pending, paid)
Money	payouts	id, worker_id, total, period, status, reference
Offers	offers	id, code, discount_type, discount_value, max_discount, min_order_value, city_ids, service_ids, vehicle_types, user_type, valid_from, valid_to, valid_days, total_usage_limit, per_user_limit, auto_apply, is_active, created_by
Offers	offer_usages	id, offer_id, user_id, booking_id, discount_amount, used_at

Booking, assignment and payments

Booking statuses (each change logged with who made it and when):

Status	Meaning	Next
PENDING_PAYMENT	Booking created, slot held for 10 minutes	CONFIRMED, or EXPIRED if unpaid
CONFIRMED	Payment verified by webhook	ASSIGNED, or CANCELLED
ASSIGNED	Offered to a worker	ACCEPTED, or REJECTED → next worker
ACCEPTED	Worker took the job	ON_THE_WAY
ON_THE_WAY	Worker travelling; live location shared	IN_PROGRESS
IN_PROGRESS	Before photos uploaded, wash under way	COMPLETED
COMPLETED	After photos uploaded	Customer rates; worker earning recorded
CANCELLED	Cancelled by customer or admin	Refund per policy

Worker assignment: Phase 1 the admin assigns manually. Phase 2 the system finds online workers in the booking's area who are free in that slot, sorted by distance and rating, and offers the job to each in turn with a 5-minute accept window. Slot capacity equals the number of workers in the area, so the business never sells more slots than it can serve.

Payment flow:

1. Customer clicks Pay; the server creates the booking (PENDING_PAYMENT), holds the slot and creates a gateway order for the final amount after discount.
2. Customer pays in the Razorpay or Stripe checkout (UPI, card, wallet).
3. The gateway calls the server's webhook; the server verifies the signature, marks the payment successful and the booking CONFIRMED. A booking is never confirmed from the browser alone.
4. The gateway settles the money into the business bank account (typically T+2 days).
5. On COMPLETED, the server adds a worker_earnings row (a commission percentage or fixed amount per wash).
6. Weekly, the admin runs payouts: bank transfer or UPI, or automatic through Razorpay Route or Stripe Connect.

7. Cancellations are refunded through the gateway's refund API according to the cancellation window.

Notifications

Event	Customer	Worker	Admin
Booking confirmed	SMS/WhatsApp + email	–	Dashboard
Worker assigned	Worker name and photo	Push notification	–
On the way / completed	SMS/WhatsApp	–	–
1 hour before slot	Reminder	Reminder	–
No worker accepted	–	–	Alert
City activated	Waitlist message	–	–

Security and compliance

- Phone OTP login with JWT tokens; role checked on every API call.
- Workers see only their own assigned jobs; customer phone numbers masked.
- Prices, discounts and payment status decided only on the server; payment webhooks verified by signature.
- HTTPS everywhere; worker KYC (ID proof, police verification) before activation.
- Terms, privacy policy and refund policy pages published (payment gateways require them for approval).

Roadmap

1. **Phase 1, MVP (about 6–8 weeks):** cities and service areas, services and city pricing, slots, customer booking with online payment and webhooks, worker job list and status updates with photos, admin manual assignment, admin-created offers, SMS notifications.
2. **Phase 2:** auto-assignment, live tracking, ratings, worker earnings and weekly payouts, reports, waitlist messaging, sub-admin roles.
3. **Phase 3:** subscription plans, referral program, native mobile apps, more cities or franchises.

Open decisions

- ☐ Country and gateway: India (₹, Razorpay) or UK (£, Stripe)?
- ☐ First launch city and its service areas
- ☐ Workers paid per job (commission) or salaried?
- ☐ Cash on delivery allowed? If yes, cash is deducted from worker payouts.
- ☐ Cancellation and refund policy (e.g. full refund if cancelled 2+ hours before the slot)